

Complex coordinates forced by finite orthogonality hypergraphs in dimension three

Mirko Navara 

*Faculty of Electrical Engineering, Czech Technical University in Prague,
Technická 2, CZ-166 27 Prague 6, Czech Republic**

Karl Svozil 

*Institute for Theoretical Physics, TU Wien, Wiedner Hauptstraße 8-10/136, A-1040 Vienna, Austria[†]
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We exhibit two finite orthogonality hypergraphs that admit faithful orthogonal representations by rays in complex three-dimensional space but admit none in real three-dimensional space. Thus, although each orthogonality diagram is unchanged, its coordinatizability in dimension three depends on whether the scalar field is complex or real. The first hypergraph has 20 vertices and 12 contexts. The second hypergraph adds one vertex and two contexts. Phase-adjusted realification also gives explicit faithful orthogonal representations of both hypergraphs in real six-dimensional space. The forced projector identities are interpreted as operational equivalences between preparations and measurement effects. We relate the obstructions to a common real structure, distinguish ray realification from probability-preserving operator realification, and extract a four-ray transition-probability witness from the displayed complex coordinates.

I. INTRODUCTION

A. Physical motivation

Can a finite collection of sharp tests on a three-level quantum system require complex coordinates? Each individual measurement basis can be chosen real, and each individual pure state can be made real by a suitable change of basis. The physical question concerns a *common* description: whether all the elementary alternatives in an overlapping family of measurements can be represented by real rays in the same three-dimensional space. A vertex shared by several contexts must denote the same test in each occurrence. The hypergraphs studied here make such a simultaneous real description impossible, even though they admit exact complex realizations.

In this interpretation, a vertex represents a rank-one projector, and a three-vertex context represents a complete projective measurement. Its outcomes are mutually exclusive and exhaustive. The distinction between $\mathbb{C}^3$ and $\mathbb{R}^3$ therefore concerns the organization of a finite family of elementary experimental alternatives, rather than the appearance of imaginary entries in a selected coordinate system. It already arises at the level of states and measurements, without specifying dynamics.

The first construction also has a preparation interpretation. Its overlapping contexts force

$$P_{a_1} + P_{a_2} = P_{b_1} + P_{b_2},$$

where the four rays are distinct and the members of each pair are nonorthogonal. Preparing either pair with equal probabilities consequently produces the same density operator when the preparation label is discarded. The two

pairs are *pseudocontexts*: their probability sums agree because of the surrounding context equations, although neither pair is a complete family of mutually exclusive outcomes [1]. Complex relative phases permit these different two-state resolutions of one mixed state; the corresponding real rank-one resolutions are rigid. In the second construction the inner pairs become orthogonal, removing this local rigidity obstruction, but the perimeter still prevents a common real realization.

These observations motivate two complementary physical readings: the freedom to resolve an unchanged mixed state into different elementary preparations, and the phase compatibility required to combine overlapping measurements into a single model. Section VII develops these readings, distinguishes preservation of an orthogonality diagram from preservation of full measurement probabilities, and identifies an operational consequence of the displayed coordinates. The mathematical results concern faithful representations by rays; experimental applications must retain the relevant assumptions about dimension, projector rank, and the identification of repeated tests.

B. Mathematical setting and relation to previous work

An orthogonality hypergraph records which finite families of rays are required to form orthogonal bases. Its representability may depend on the scalar field, even when the ambient dimension is fixed. Harding and Salinas Schmeis gave the first finite hypergraph with a faithful orthogonal representation over $\mathbb{C}^3$ but none over $\mathbb{R}^3$ [2]. Independently, Trandafir and Cabello encountered the same real-complex separation in a Kochen–Specker (KS) construction designed for bipartite perfect quantum strategies [3]. A subsequent analysis by Navara and Svozil relates and extends those construc-

* navara@fel.cvut.cz

† karl.svozil@tuwien.ac.at

tions through mutually unbiased bases and KS closures [4].

The role of complex scalars in quantum theory involves two questions that should not be conflated. If a change of dimension and an additional complex structure are allowed, a complex Hilbert space $\mathbb{C}^n$ can be represented on the real space $\mathbb{R}^{2n}$ by separating real and imaginary parts; related constructions reproduce broad classes of complex quantum protocols within a real formalism [5–7]. If instead one keeps fixed the dimension, composition rules, locality assumptions, or the interpretation of a context as a maximal orthogonal family, real and complex models can be separated operationally [8–10]. A statement that a hypergraph is representable in $\mathbb{C}^3$ but not in $\mathbb{R}^3$ belongs to the latter, fixed-dimension comparison.

The phase-adjusted realification method of Khrennikov and Svozil [11] makes the boundary precise for finite orthogonality data. It preserves and reflects all pairwise orthogonalities in $\mathbb{R}^6$, but a three-ray context then spans only half of the ambient real space. Below we apply this construction explicitly to both octagons after proving their same-dimension separation.

We present two further exact finite examples, both with an octagonal incidence skeleton. The 20-vertex hypergraph H_{20} forces an equality of two sums of rank-one projectors. Complex phases permit two distinct nonorthogonal two-ray decompositions of the same positive operator; over the reals, the spectral decomposition fixes the rays. The 21-vertex hypergraph H_{21} is obtained by completing both inner pairs with a common ray. The spectral argument then becomes degenerate, but the perimeter closure relations supply a second obstruction to real coordinates.

Unlike a conventional Kochen–Specker set, neither hypergraph is used through an uncolorability argument. Its role here is instead to enforce exact orthogonality incidences and the resulting projector or closure identities; no contradiction involving two-valued states is assumed or needed.

The results established below go beyond merely exhibiting convenient complex coordinates. For each displayed representation we verify all unordered ray pairs exactly: the prescribed pairs are orthogonal, every other inner product is nonzero, and distinct vertices label distinct rays. Thus the word “faithful” carries its full graph-theoretic meaning [12, 13].

II. ORTHOGONALITY REPRESENTATIONS AND TWO ALGEBRAIC LEMMAS

Throughout, we use the physics convention for the Hermitian inner product, which is conjugate-linear in the first argument:

$$\langle x, y \rangle = \sum_{k=1}^d \overline{x_k} y_k. \quad (1)$$

Definition 1. Let $H = (V, \mathcal{E})$ be a three-uniform hypergraph, let $\mathbb{F}$ be either $\mathbb{R}$ or $\mathbb{C}$, and let $d \geq 3$. A *faithful orthogonal representation* of H in $\mathbb{F}^d$ assigns a nonzero vector $x_v \in \mathbb{F}^d$ to every $v \in V$ such that:

1. distinct vertices determine distinct rays; and
2. $\langle x_u, x_v \rangle = 0$ if and only if u and v belong to a common hyperedge.

Each hyperedge is then an orthogonal triple. For $d = 3$ it is an orthogonal basis and hence a maximal *context*; for $d > 3$ the same specified triple need not be maximal.

For a nonzero column vector x , let $x^\dagger$ denote its conjugate transpose and write

$$P_x = \frac{xx^\dagger}{\langle x, x \rangle} \quad (2)$$

for the rank-one projector onto its ray. In a three-dimensional representation, a context $\{x, y, z\}$ satisfies

$$P_x + P_y + P_z = I_3. \quad (3)$$

Consequently, integer linear combinations of three-dimensional context equations can force projector identities that hold in every representation of the hypergraph in that dimension.

For $x, y \in \mathbb{C}^3$ define the Hermitian cross product by

$$x \boxtimes y := \overline{x \times y}, \quad (4)$$

where $x \times y$ is the ordinary bilinear cross product. Then $x \boxtimes y$ is Hermitian-orthogonal to both x and y .

Lemma 2 (Cyclic Gram closure). *Let $x, y, z \in \mathbb{C}^3$, and suppose the two cross products in Eq. (4) are nonzero. Then*

$$\langle x \boxtimes y, y \boxtimes z \rangle = 0 \iff \langle x, y \rangle \langle y, z \rangle = \langle x, z \rangle \langle y, y \rangle. \quad (5)$$

Proof. The bilinear Lagrange identity, with $\cdot$ denoting the standard bilinear dot product on $\mathbb{C}^3$, gives

$$(x \times y) \cdot (\bar{y} \times \bar{z}) = (x \cdot \bar{y})(y \cdot \bar{z}) - (x \cdot \bar{z})(y \cdot \bar{y}).$$

Indeed, for arbitrary $a, b \in \mathbb{C}^3$, $\langle \bar{a}, \bar{b} \rangle = a \cdot \bar{b}$; taking $a = x \times y$ and $b = y \times z$ therefore identifies the left-hand side with $\langle x \boxtimes y, y \boxtimes z \rangle$. The right-hand side is the complex conjugate of

$$\langle x, y \rangle \langle y, z \rangle - \langle x, z \rangle \langle y, y \rangle.$$

Their vanishing is therefore equivalent. $\square$

Lemma 3 (Rigidity of two real rays). *Let a_1, a_2, b_1, b_2 be real unit vectors satisfying*

$$P_{a_1} + P_{a_2} = P_{b_1} + P_{b_2}. \quad (6)$$

If a_1 and a_2 are distinct and nonorthogonal, then $\{[b_1], [b_2]\} = \{[a_1], [a_2]\}$ as unordered pairs of rays.

Proof. All four vectors lie in the range of the common operator in Eq. (6). Choose signs so that $\gamma = \langle a_1, a_2 \rangle \in (0, 1)$. The two nonzero eigenvalues are $1 + \gamma$ and $1 - \gamma$, with eigenvectors $a_1 + a_2$ and $a_1 - a_2$, respectively. Choose signs and order for the b -vectors so that $\delta = \langle b_1, b_2 \rangle \geq 0$. Equality of spectra gives $\delta = \gamma$. Since the two eigenvalues are distinct, there are real nonzero scalars λ, μ such that

$$b_1 + b_2 = \lambda(a_1 + a_2), \quad b_1 - b_2 = \mu(a_1 - a_2).$$

The unit-norm and inner-product equations give

$$\begin{aligned} \lambda^2(1 + \gamma) + \mu^2(1 - \gamma) &= 2, \\ \lambda^2(1 + \gamma) - \mu^2(1 - \gamma) &= 2\gamma. \end{aligned}$$

Hence $\lambda^2 = \mu^2 = 1$. Solving for b_1, b_2 recovers the two original rays, possibly with signs and permutation. $\square$

III. THE 20-RAY OCTAGON

The vertices of H_{20} are eight midpoint vertices $v_0, \dots, v_7$, eight corner vertices $v_{01}, v_{12}, \dots, v_{70}$, and four inner vertices a_1, a_2, b_1, b_2 . Indices are read modulo eight. The eight perimeter contexts and four inner contexts are

$$E_j = \{v_{j-1,j}, v_j, v_{j,j+1}\}, \quad j = 0, \dots, 7, \quad (7)$$

$$\begin{aligned} D_0 &= \{v_0, a_1, v_4\}, & D_2 &= \{v_2, a_2, v_6\}, \\ D_1 &= \{v_1, b_1, v_5\}, & D_3 &= \{v_3, b_1, v_7\}. \end{aligned} \quad (8)$$

The incidence structure is shown in Fig. 1.

A. The covering identity

Consider the two collections of six contexts

$$\begin{aligned} \mathcal{A} &= \{D_0, D_2, E_1, E_3, E_5, E_7\}, \\ \mathcal{B} &= \{D_1, D_3, E_0, E_2, E_4, E_6\}. \end{aligned}$$

Every midpoint and every corner occurs exactly once in each collection. Only the two a -vertices occur in $\mathcal{A}$, and only the two b -vertices occur in $\mathcal{B}$. Summing Eq. (3) over the two collections and subtracting therefore gives

$$P_{a_1} + P_{a_2} = P_{b_1} + P_{b_2}. \quad (9)$$

This identity is combinatorial: it holds in every real or complex three-dimensional representation of H_{20} .

B. An exact faithful representation over $\mathbb{C}^3$

Set

$$\begin{aligned} C &= \frac{49 - 5\sqrt{69}}{26}, & s &= \sqrt{C}, \\ N &= 1 + C, & r &= \sqrt{N}. \end{aligned} \quad (10)$$

The number C is the smaller positive root of

$$13C^2 - 49C + 13 = 0, \quad (11)$$

so $0 < C < 1$. Define the unit phases

$$\eta = \frac{5 + 12i}{13}, \quad (12)$$

$$\zeta = \frac{14 + 6\sqrt{69}}{65} + i\frac{4\sqrt{69} - 21}{65}. \quad (13)$$

Let

$$\begin{aligned} (\mu_0, \dots, \mu_7) &= (-1, \eta, 1, -\eta, -1, \eta, 1, -\eta), \\ (\tau_0, \dots, \tau_7) &= (1, \zeta, -i, -i\zeta, -1, -\zeta, i, i\zeta). \end{aligned}$$

The inner and midpoint rays are represented by

$$\begin{aligned} a_1 &= (s, 1, 0), & a_2 &= (s, -1, 0), \\ b_1 &= (s, \eta, 0), & b_2 &= (s, -\eta, 0), \end{aligned} \quad (14)$$

$$v_j = (1, s\mu_j, r\tau_j), \quad j = 0, \dots, 7. \quad (15)$$

For j modulo eight, define

$$v_{j,j+1} = v_j \boxtimes v_{j+1}. \quad (16)$$

Equivalently, Eq. (16) is the explicit formula

$$\begin{aligned} v_{j,j+1} &= (sr(\bar{\mu}_j\bar{\tau}_{j+1} - \bar{\tau}_j\bar{\mu}_{j+1}), \\ &\quad r(\bar{\tau}_j - \bar{\tau}_{j+1}), s(\bar{\mu}_{j+1} - \bar{\mu}_j)). \end{aligned} \quad (17)$$

Theorem 4. Equations (10)–(17) give a faithful orthogonal representation of H_{20} in $\mathbb{C}^3$.

Proof. First, $|\eta| = 1$. Write $\zeta = x + iy$. The four inner contexts follow directly from $r^2 = N$, $|\eta| = |\zeta| = 1$, and the antipodal relations $\mu_{j+4} = \mu_j$, $\tau_{j+4} = -\tau_j$. For example,

$$\langle a_1, v_0 \rangle = 0, \quad \langle v_0, v_4 \rangle = 1 + C - r^2 = 0,$$

and

$$\langle b_2, v_1 \rangle = 0, \quad \langle v_1, v_5 \rangle = 1 + C - r^2|\zeta|^2 = 0.$$

The other two inner contexts are identical calculations.

By construction, $v_{j,j+1}$ is orthogonal to v_j and v_{j+1} . It remains to prove orthogonality of the two corner rays in every perimeter context. Put

$$G_{jk} = \langle v_j, v_k \rangle = 1 + C\bar{\mu}_j\mu_k + N\bar{\tau}_j\tau_k.$$

Lemma 2 reduces the eight remaining conditions to

$$G_{j-1,j}G_{j,j+1} = G_{j-1,j+1}G_{j,j}, \quad j = 0, \dots, 7. \quad (18)$$

Substituting the cyclic patterns for μ_j and τ_j reduces all eight conditions to common real and imaginary equations once $x^2 + y^2 = 1$. Separating those parts gives

$$7Cx - 17Cy - 13C - 13x + 13y + 13 = 0, \quad (19)$$

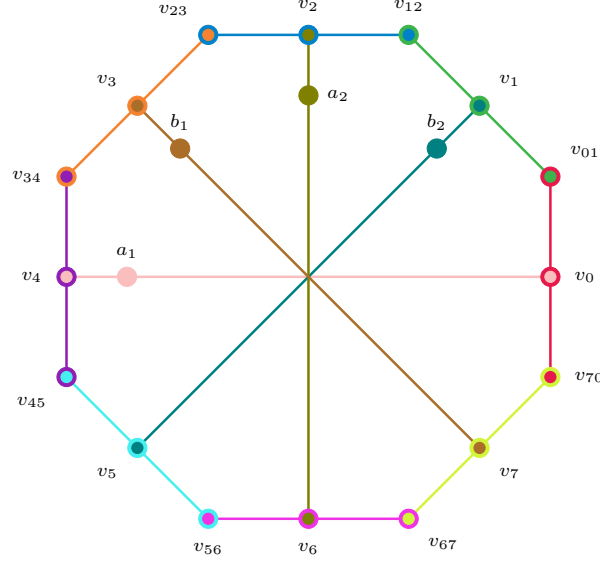


FIG. 1. The 20-ray hypergraph H_{20} . The eight perimeter contexts $E_0, \dots, E_7$ form the octagon; the four inner contexts join opposite midpoints through a_1, a_2, b_1, b_2 . Each color denotes one context.

and

$$-7C^2x + 17C^2y + 13C^2 - 20Cx + 30Cy + 2C - 13x + 13y + 13 = 0. \quad (20)$$

Solving these two linear equations for x and y gives

$$x = \frac{104 - 76C}{65(1 + C)}, \quad y = \frac{39 - 81C}{65(1 + C)}. \quad (21)$$

For these values,

$$x^2 + y^2 - 1 = \frac{48(13C^2 - 49C + 13)}{325(1 + C)^2} = 0. \quad (22)$$

Using the smaller root in Eq. (11), Eq. (21) is exactly the phase in Eq. (13). Thus all twelve triples are contexts.

It remains to exclude unintended orthogonalities and ray coincidences. The exact overlap certificate in Appendix A shows that the 36 prescribed unordered pairs have zero overlap, whereas the 154 unprescribed pairs have squared normalized overlaps between $1/25$ and $4/5$. Hence every unprescribed inner product is nonzero and no two displayed vectors are proportional. The representation is faithful. $\square$

The two inner pairs indeed decompose the same operator. Since all four vectors have squared norm N and $|\eta| = 1$,

$$P_{a_1} + P_{a_2} = P_{b_1} + P_{b_2} = \frac{2}{N} \text{diag}(C, 1, 0). \quad (23)$$

Their within-pair inner products are $C - 1 \neq 0$, and the four rays are distinct because $\eta \notin \{1, -1\}$.

C. Nonexistence over $\mathbb{R}^3$

Theorem 5. *The hypergraph H_{20} has no faithful orthogonal representation in $\mathbb{R}^3$.*

Proof. Suppose a faithful real representation existed. The covering argument is independent of the field, so Eq. (9) would hold. Faithfulness implies that a_1, a_2 are distinct and nonorthogonal, because they do not share a context; the same is true of b_1, b_2 . Lemma 3 then identifies the unordered pair of b -rays with the unordered pair of a -rays. This contradicts the requirement that four distinct vertices label four distinct rays. $\square$

IV. THE 21-RAY ORTHOGONAL AUGMENTATION

Add one vertex c and two contexts

$$F_a = \{a_1, c, a_2\}, \quad F_b = \{b_1, c, b_2\} \quad (24)$$

to H_{20} . The result is the 21-vertex, 14-context hypergraph H_{21} shown in Fig. 2. The added contexts force both inner pairs to be orthogonal and to have the same one-dimensional orthogonal complement.

A. Exact coordinates over $\mathbb{C}^3$

Let

$$p = \frac{\sqrt{14} + \sqrt{2}}{4}, \quad q = \frac{\sqrt{14} - \sqrt{2}}{4}. \quad (25)$$

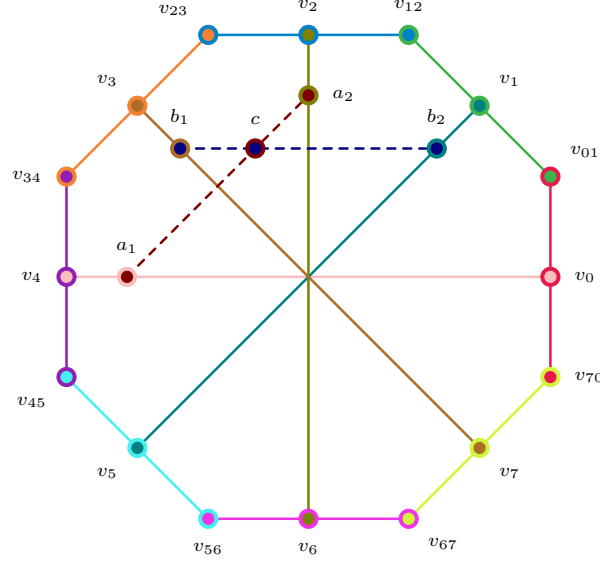


FIG. 2. The 21-ray hypergraph H_{21} . The twelve solid contexts are those of H_{20} ; the two dashed contexts meet at the new vertex c .

Then

$$p^2 + q^2 = 2, \quad pq = \frac{3}{4}. \quad (26)$$

Choose

$$\begin{aligned} c &= (0, 0, 1), \\ a_1 &= (1, 1, 0), & a_2 &= (1, -1, 0), \\ b_1 &= (1, i, 0), & b_2 &= (1, -i, 0), \end{aligned} \quad (27)$$

and the eight midpoint representatives

$$\begin{aligned} v_0 &= (1, -1, \sqrt{2}), & v_4 &= (1, -1, -\sqrt{2}), \\ v_1 &= (1, i, p + iq), & v_5 &= (1, i, -p - iq), \\ v_2 &= (1, 1, -i\sqrt{2}), & v_6 &= (1, 1, i\sqrt{2}), \\ v_3 &= (1, -i, q - ip), & v_7 &= (1, -i, -q + ip). \end{aligned} \quad (28)$$

The eight corner representatives are

$$\begin{aligned} v_{01} &= (-p + i(q + \sqrt{2}), \sqrt{2} - p + iq, 1 - i), \\ v_{12} &= (\sqrt{2} - p + iq, p - i(q + \sqrt{2}), 1 + i), \\ v_{23} &= (q + \sqrt{2} + ip, -q + i(\sqrt{2} - p), -1 + i), \\ v_{34} &= (q + i(p - \sqrt{2}), q + \sqrt{2} + ip, -1 - i), \\ v_{45} &= (p - i(q + \sqrt{2}), -\sqrt{2} + p - iq, 1 - i), \\ v_{56} &= (-\sqrt{2} + p - iq, -p + i(q + \sqrt{2}), 1 + i), \\ v_{67} &= (-q - \sqrt{2} - ip, q - i(\sqrt{2} - p), -1 + i), \\ v_{70} &= (-q - i(p - \sqrt{2}), -q - \sqrt{2} - ip, -1 - i). \end{aligned} \quad (29)$$

Theorem 6. Equations (25)–(29) give a faithful orthogonal representation of H_{21} in $\mathbb{C}^3$.

Proof. The two added contexts are immediate from Eq. (27). For the original inner contexts, representative calculations are

$$\begin{aligned} \langle a_1, v_0 \rangle &= 0, & \langle v_0, v_4 \rangle &= 1 + 1 - 2 = 0, \\ \langle b_2, v_1 \rangle &= 0, & \langle v_1, v_5 \rangle &= 2 - (p^2 + q^2) = 0. \end{aligned}$$

The two remaining inner contexts follow in the same way.

Direct evaluation of the Hermitian cross product gives, exactly,

$$v_{j,j+1} = v_j \boxtimes v_{j+1}$$

for each representative in Eq. (29). Hence every corner is orthogonal to its two incident midpoints. Substitution in the Gram closure identity (5) reduces the eight remaining perimeter inner products to the two relations in Eq. (26); they therefore vanish.

Finally, Appendix A gives an exact check of all 210 unordered pairs. The 42 pairs prescribed by the fourteen contexts have zero overlap. Every remaining pair has squared normalized overlap either $1/4$ or $1/2$. Thus there are no additional orthogonalities and no ray coincidences. $\square$

B. Nonexistence over $\mathbb{R}^3$

Theorem 7. The hypergraph H_{21} has no faithful orthogonal representation in $\mathbb{R}^3$.

Proof. Assume that a faithful real representation exists. A global orthogonal transformation and independent rescalings of ray representatives allow us to set

$$c = (0, 0, 1), \quad a_1 = (1, 0, 0), \quad a_2 = (0, 1, 0). \quad (30)$$

The second added context lies in the plane $c^\perp$, so for some angle θ ,

$$b_1 = (\cos \theta, \sin \theta, 0), \quad b_2 = (-\sin \theta, \cos \theta, 0). \quad (31)$$

Faithfulness excludes $\theta \in \frac{\pi}{2}\mathbb{Z}$, since those values identify a b -ray with an a -ray. Put

$$m = \tan \theta \neq 0, \quad M = 1 + m^2. \quad (32)$$

Write a representative vector of a midpoint ray as $v_j = (X_j, Y_j, Z_j)$. No midpoint ray can lie in $c^\perp$. Indeed, a planar vector orthogonal to the inner ray in its context is proportional to the other member of one of the two planar bases $\{a_1, a_2\}$ or $\{b_1, b_2\}$, contradicting faithfulness. Thus $Z_j \neq 0$ for every j , and each representative may be rescaled to

$$v_j = (X_j, Y_j, 1). \quad (33)$$

The inner contexts D_0, D_1, D_3 imply

$$X_0 = X_4 = 0, \quad Y_0 Y_4 + 1 = 0, \quad (34)$$

$$Y_1 = mX_1, \quad Y_5 = mX_5, \quad MX_1 X_5 + 1 = 0, \quad (35)$$

$$X_3 = -mY_3, \quad X_7 = -mY_7, \quad MY_3 Y_7 + 1 = 0. \quad (36)$$

In particular,

$$X_5 = -\frac{1}{MX_1}, \quad Y_7 = -\frac{1}{MY_3}, \quad (37)$$

and $X_1 Y_3 \neq 0$.

The corner rays can be eliminated from the perimeter by the real form of Lemma 2. At $j = 0$, Eq. (5) factors as

$$Y_0((mX_1 Y_7 - 1)Y_0 + Y_7 + mX_1) = 0. \quad (38)$$

The first factor cannot vanish because of Eq. (34). Moreover, the coefficient $1 - mX_1 Y_7$ cannot vanish: otherwise Eq. (38) would also give $Y_7 + mX_1 = 0$, and the two relations would imply $-m^2 X_1^2 = 1$. Therefore

$$Y_0 = \frac{Y_7 + mX_1}{1 - mX_1 Y_7}. \quad (39)$$

The closure equation at $j = 4$ gives analogously

$$Y_4 = \frac{Y_3 + mX_5}{1 - mY_3 X_5}, \quad (40)$$

with a nonzero denominator.

Substituting Eq. (37) into Eqs. (39) and (40) yields

$$Y_0 = \frac{mMX_1 Y_3 - 1}{MY_3 + mX_1}, \quad Y_4 = \frac{MX_1 Y_3 - m}{MX_1 + mY_3}. \quad (41)$$

Set $u = X_1 Y_3$. The remaining equation $Y_0 Y_4 = -1$ becomes

$$(mMu - 1)(Mu - m) = -(mX_1 + MY_3)(MX_1 + mY_3). \quad (42)$$

Expanding the left-hand side gives

$$\begin{aligned} (mMu - 1)(Mu - m) &= mM^2 u^2 - M(m^2 + 1)u + m \\ &= mM^2 u^2 - M^2 u + m, \end{aligned}$$

where the second equality uses $M = 1 + m^2$. The right-hand side is

$$-u(M^2 + m^2) - mM(X_1^2 + Y_3^2).$$

After cancellation and division by $m \neq 0$, Eq. (42) is therefore equivalent to the master equation

$$M^2 u^2 + mu + 1 + M(X_1^2 + Y_3^2) = 0. \quad (43)$$

Since X_1, Y_3 are real, $X_1^2 + Y_3^2 \geq 2|X_1 Y_3| = 2|u|$. The left-hand side of Eq. (43) is consequently at least

$$\begin{aligned} M^2 u^2 + mu + 1 + 2M|u| &\geq 1 + M^2 u^2 + |u|(2M - |m|) \\ &\geq 1. \end{aligned}$$

Here, with $t = |m|$,

$$2M - |m| = 2t^2 - t + 2 = 2\left(t - \frac{1}{4}\right)^2 + \frac{15}{8} > 0.$$

This contradicts Eq. (43), and the assumed real representation cannot exist. $\square$

V. FAITHFUL REALIFICATION IN SIX DIMENSIONS

Theorems 5 and 7 keep the ambient dimension fixed. When dimension doubling is allowed, both hypergraphs have faithful real representations. We first recall the finite construction and then give explicit phase choices for the present coordinates [11].

Theorem 8 (Phase-adjusted realification). *Every finite faithful orthogonal representation by rays in $\mathbb{C}^3$ induces a faithful orthogonal representation of the same hypergraph by rays in $\mathbb{R}^6$.*

Proof. Let $\psi_1, \dots, \psi_N \in \mathbb{C}^3$ be nonzero representatives and put $c_{k\ell} = \langle \psi_k, \psi_\ell \rangle$. Define the canonical realification

$$\Phi_0(z_1, z_2, z_3) = (\Re z_1, \Re z_2, \Re z_3, \Im z_1, \Im z_2, \Im z_3). \quad (44)$$

For phases $\theta_k \in \mathbb{R}$, set $R_k = \Phi_0(e^{i\theta_k} \psi_k)$. Direct expansion gives

$$R_k \cdot R_\ell = \Re(e^{i(\theta_\ell - \theta_k)} c_{k\ell}). \quad (45)$$

Thus $c_{k\ell} = 0$ implies $R_k \cdot R_\ell = 0$ for every phase choice. If $c_{k\ell} \neq 0$, write $c_{k\ell} = |c_{k\ell}|e^{i\varphi_{k\ell}}$. The right-hand side of Eq. (45) vanishes precisely when

$$\theta_\ell - \theta_k + \varphi_{k\ell} \equiv \frac{\pi}{2} \pmod{\pi}. \quad (46)$$

Choose $\theta_1, \dots, \theta_N$ inductively. Suppose that the first $m - 1$ phases have been fixed. For each $k < m$ with $c_{km} \neq 0$, the unwanted orthogonality equation $R_k \cdot R_m = 0$ excludes, by Eq. (46), exactly two values of θ_m modulo 2π . Their union is finite, so a phase outside it exists. Continuing through $m = N$ makes the real dot product nonzero for every originally nonorthogonal pair, while all original orthogonalities remain zero.

It remains only to check ray distinctness. If R_k and R_ℓ were real proportional, then $e^{i\theta_k}\psi_k$ and $e^{i\theta_\ell}\psi_\ell$ would be real proportional and hence would represent the same complex ray. Faithfulness of the original representation excludes this. Therefore the realified representation is faithful. $\square$

For explicit coordinates it is convenient to parametrize the phase by a real number t :

$$\lambda_t = \frac{1 + it}{\sqrt{1 + t^2}}, \quad \Phi_t(z) = \Phi_0(\lambda_t z). \quad (47)$$

Writing $z = x + iy$ with $x, y \in \mathbb{R}^3$ gives the entirely real formula

$$\Phi_t(x + iy) = \frac{(x - ty, y + tx)}{\sqrt{1 + t^2}} \in \mathbb{R}^6. \quad (48)$$

Moreover,

$$\Phi_{t_u}(u) \cdot \Phi_{t_v}(v) = \Re \left(\frac{(1 - it_u)(1 + it_v)}{\sqrt{(1 + t_u^2)(1 + t_v^2)}} \langle u, v \rangle \right). \quad (49)$$

Corollary 9 (Explicit real six-dimensional coordinates). *Let $\psi_v^{(20)}$ denote the representatives in Eqs. (10)–(17). Define*

$$t_v^{(20)} = \begin{cases} 1, & v \in \{v_{34}, v_{45}, v_{56}\}, \\ 2, & v \in \{v_{67}, v_{70}\}, \\ 0, & \text{otherwise.} \end{cases} \quad (50)$$

Then $R_v^{(20)} = \Phi_{t_v^{(20)}}(\psi_v^{(20)})$ is a faithful orthogonal representation of H_{20} in $\mathbb{R}^6$.

Likewise, let $\psi_v^{(21)}$ denote the representatives in Eqs. (25)–(29), and define

$$t_v^{(21)} = \begin{cases} 1, & v = v_{45}, \\ 2, & v \in \{v_2, v_3, v_6, v_7, v_{34}, v_{56}, v_{70}\}, \\ 3, & v = v_{67}, \\ 0, & \text{otherwise.} \end{cases} \quad (51)$$

Then $R_v^{(21)} = \Phi_{t_v^{(21)}}(\psi_v^{(21)})$ is a faithful orthogonal representation of H_{21} in $\mathbb{R}^6$.

Proof. Equations (48), (50), and (51) give all six real coordinates directly from the complex vectors already displayed. Exact substitution in Eq. (49) gives zero for exactly 36 of the $\binom{20}{2} = 190$ pairs in H_{20} , namely the pairs

prescribed by its twelve contexts. The other 154 real dot products are nonzero. For H_{21} , exactly 42 of the $\binom{21}{2} = 210$ real dot products vanish, namely the pairs prescribed by its fourteen contexts, and the other 168 are nonzero. These calculations take place in the algebraic fields generated by the radicals in the complex coordinates and by $\sqrt{1 + t^2}$; no numerical tolerance is involved. Ray distinctness follows from Theorem 8. $\square$

Each displayed three-ray context is therefore an orthogonal triple in $\mathbb{R}^6$, but it spans only a three-dimensional subspace. Its projector sum is the rank-three projector onto that subspace, not I_6 . This loss of maximality is precisely why the real six-dimensional representations do not contradict the two real three-dimensional no-go theorems.

VI. DISCUSSION

The two nonexistence proofs expose different real-complex mechanisms. For H_{20} , the context covering forces a rank-two positive operator to have two disjoint, nonorthogonal decompositions into two rank-one projectors. Complex relative phases make this possible. Over the reals, the two simple eigenvalues determine the two rays, producing the rigidity of Lemma 3.

For H_{21} , both inner pairs are orthogonal and their common projector sum is the identity on $c^\perp$. The eigenvalue argument is therefore unavailable: the relevant eigenvalue is degenerate and many real orthogonal bases of $c^\perp$ exist. The obstruction instead comes from compatibility with the octagonal perimeter. The two reduced closure equations determine Y_0 and Y_4 , while the inner context D_0 demands $Y_0 Y_4 = -1$; together they produce the positive master expression in Eq. (43).

Both statements are dimension-specific. Corollary 9 realizes the same pairwise orthogonality relations faithfully in $\mathbb{R}^6$, but the specified triples are no longer complete bases. Consequently, the three-dimensional context equation (3) is replaced by a rank-three subspace projector. The complementary covering no longer forces Eq. (9), and the three-dimensional cross-product closure used for H_{21} no longer applies. The comparison therefore concerns the scalar field at fixed dimension and fixed context maximality, not the possibility of simulating complex coordinates in a larger real space. It is also independent of KS noncolorability: the contradictions use orthogonality, completeness of contexts, and field-dependent algebra, not an impossibility of assigning two-valued states.

These fixed-dimension witnesses may also guide the design of protocols in which a real implementation must either enlarge the Hilbert-space dimension or relinquish maximality of the specified contexts.

Finally, faithfulness is essential. If unprescribed orthogonalities or ray coincidences were allowed, the covering identity could collapse to a degenerate realization.

Conversely, deleting contexts weakens the projector identity or the perimeter closure system. The relevant object is therefore the complete pair consisting of the vertex set and its specified contexts.

VII. PHYSICAL INTERPRETATION AND OPERATIONAL PROSPECTS

A. Pseudocontexts as equivalent preparations and measurement effects

The covering identity (9) has two operational interpretations. First, for every density operator ρ ,

$$\text{tr}(\rho(P_{a_1} + P_{a_2})) = \text{tr}(\rho(P_{b_1} + P_{b_2})).$$

This equality is state independent, whereas the common value generally depends on ρ and need not equal one. The two nonorthogonal pairs are therefore not two-outcome projective measurements. More generally, the same complementary-cover argument applies to any single assignment $p(v)$ satisfying $\sum_{v \in E} p(v) = 1$ for every context, independently of the Born rule [1]. The equality alone does not distinguish real and complex quantum theory; the field dependence enters through its faithful realization by rank-one projectors.

Second, the normalized common operator is a density operator,

$$\rho_* := \frac{1}{2}(P_{a_1} + P_{a_2}) = \frac{1}{2}(P_{b_1} + P_{b_2}).$$

Thus two different equal-probability preparations become indistinguishable by every subsequent measurement on the prepared system when their labels are forgotten. For the coordinates of Sec. III, this freedom can be displayed explicitly:

$$\psi_{\pm}(\phi) = \frac{(\sqrt{C}, \pm e^{i\phi}, 0)}{\sqrt{1+C}}, \quad \rho_* = \frac{\text{diag}(C, 1, 0)}{1+C}.$$

Every value of ϕ gives the same equal-weight mixture. The inner pairs in Eq. (14) correspond to $\phi = 0$ and $e^{i\phi} = \eta$. This phase family describes the decompositions of the inner operator; it does not assert that every phase extends to a faithful realization of the entire octagon.

On the Bloch sphere of the common two-dimensional support, an equal-weight decomposition is a chord with midpoint ρ_* . Since $0 < C < 1$, that midpoint is not the center. The complex sphere admits a circle of chord directions with this midpoint. In a fixed real-state disk containing ρ_* , only one unordered pair remains. Lemma 3 is the algebraic version of this restriction. The freedom detected by H_{20} is consequently more specific than the general nonuniqueness of mixed-state ensembles: it requires distinct *equal-weight two-state* decompositions of a nonmaximally mixed state on its support.

There is also a measurement realization. With $S = P_{a_1} + P_{a_2} = P_{b_1} + P_{b_2}$, the two families

$$\{\tfrac{1}{2}P_{a_1}, \tfrac{1}{2}P_{a_2}, I - \tfrac{1}{2}S\}, \quad \{\tfrac{1}{2}P_{b_1}, \tfrac{1}{2}P_{b_2}, I - \tfrac{1}{2}S\}$$

are positive-operator-valued measures: their effects are positive and sum to the identity. They provide different resolutions of the same coarse-grained effect $S/2$ and have the same remaining outcome.

Finally, the ensemble interpretation connects to remote preparation [14]. The purification

$$|\Psi\rangle = \frac{\sqrt{C}|0\rangle|0\rangle + |1\rangle|1\rangle}{\sqrt{1+C}}$$

has reduced state ρ_* on the first system. Measuring the second system in the basis $(|0\rangle \pm e^{-i\phi}|1\rangle)/\sqrt{2}$ prepares $\psi_{\pm}(\phi)$, each with probability $1/2$. This supplies a concrete way to generate the two pseudocontext ensembles without changing the unconditioned reduced state. A steering or communication advantage from the full hypergraph would require an additional protocol and a separate performance bound.

B. A common real structure and relative phase

The physical content of the nonexistence theorems is invariant under a common unitary transformation and independent choices of ray phases. Equivalently, no faithful realization of either hypergraph in $\mathbb{C}^3$ admits an antiunitary involution K , with $K^2 = I$, that fixes every labelled projector: $KP_vK^{-1} = P_v$ for all v . Such an involution would define a conjugation, and a basis of its fixed real subspace would make all the projectors real. Each invariant ray has a representative fixed by K , so this would contradict the real three-dimensional no-go theorem.

For H_{21} , the two inner orthogonal pairs by themselves admit a common real description. Their equal mixtures are maximally mixed on $c^{\perp}$, where real orthogonal decompositions are no longer unique. The impossibility appears when these pairs must also fit the entire perimeter. This provides a global phase compatibility interpretation of the closure obstruction. It does not identify a dynamical geometric phase around the drawn octagon: inner products of orthogonal rays vanish and have no defined phase.

A phase-sensitive quantity can instead be formed from nonorthogonal rays. For unit representatives, the Bargmann invariant

$$B(x, y, z) := \text{tr}(P_x P_y P_z) = \langle x, y \rangle \langle y, z \rangle \langle z, x \rangle$$

is invariant under ray rephasings and a common unitary transformation. A nonreal value excludes a simultaneous real representation of those projectors; such cyclic overlaps connect geometric phase to witnesses of quantum imaginarity [15, 16]. In the displayed H_{20} representation, direct substitution yields

$$B(a_1, a_2, b_1) = \frac{(C-1)(C-\eta)(C+\bar{\eta})}{(1+C)^3} = \frac{115 + 8i\sqrt{69}}{375}.$$

The nonzero imaginary part makes the irreducible relative phase explicit. This particular value is a property of

the supplied coordinates, not a claimed invariant value of every faithful representation of the hypergraph. Complex conjugation preserves all orthogonalities and pairwise transition probabilities while conjugating B . The incidence structure therefore does not select a handedness of phase, and no statement about physical time-reversal violation follows.

C. What a real simulation preserves

Two kinds of realification have different operational meanings. The phase-adjusted construction in Sec. V assigns a *real rank-one* projector to each vertex and preserves exactly the orthogonality relations. It need not preserve nonzero transition probabilities or the pseudo-context projector identity. Each specified triple is incomplete in the real six-dimensional space.

By contrast, write a complex operator as $A + iB$, with real matrices A, B , and use the standard operator representation

$$\mathcal{R}(A + iB) = \begin{pmatrix} A & -B \\ B & A \end{pmatrix}.$$

For a density operator and a projector, respectively, set $\rho_{\mathbb{R}} = \mathcal{R}(\rho)/2$ and $\Pi_v = \mathcal{R}(P_v)$. Then

$$\text{tr}_{\mathbb{R}}(\rho_{\mathbb{R}}\Pi_v) = \text{tr}_{\mathbb{C}}(\rho P_v), \quad \sum_{v \in E} \Pi_v = I_6.$$

This construction preserves probabilities, context completeness, and the pseudocontext identity, but each complex rank-one projector becomes a *real rank-two* projector. The three outcomes remain a complete projective measurement, with each outcome now a two-dimensional subspace of the real model. This distinction is consistent with the broader real simulation constructions cited in the Introduction.

The rank change is essential if the nondegenerate two-projector identity is retained. The proof of Lemma 3 applies in any ambient real dimension, because equality of the two positive sums confines all four rays to the same rank-two support. Enlarging dimension alone therefore cannot realize that identity with four distinct real rank-one projectors and nonorthogonal members within each pair. Dimension three is needed to *derive* the identity from the complete contexts of H_{20} ; the rigidity of the resulting identity is dimension independent.

D. An operational consequence and experimental scope

The displayed coordinates also give a compact transition-probability consequence. Regard a label u as the pure preparation P_u , and a label v as the rank-one yes-no test $\{P_v, I - P_v\}$. The response probability is

$Q(u, v) = \text{tr}(P_u P_v)$, as in Eq. (A1). For the four inner rays of H_{20} ,

$$\begin{aligned} Q(a_1, a_2) &= Q(b_1, b_2) = \frac{23}{75}, \\ Q(a_1, b_1) &= Q(a_2, b_2) = \frac{59}{75}, \\ Q(a_1, b_2) &= Q(a_2, b_1) = \frac{13}{25}. \end{aligned}$$

These values follow directly from Eq. (14) and $13C^2 - 49C + 13 = 0$. They cannot be reproduced by real unit vectors in any dimension. To see this, every three-ray subset has the sum of its three squared overlaps equal to $121/75 > 1$. For real inner products x, y, z , positivity of the corresponding Gram determinant requires $1 - x^2 - y^2 - z^2 + 2xyz \geq 0$, hence $xyz > 0$. Changing representative signs can therefore make all six inner products positive. In the order (a_1, a_2, b_1, b_2) the real Gram matrix would be

$$G = \begin{pmatrix} 1 & g & h & \ell \\ g & 1 & \ell & h \\ h & \ell & 1 & g \\ \ell & h & g & 1 \end{pmatrix},$$

where

$$g = \sqrt{\frac{23}{75}}, \quad h = \sqrt{\frac{59}{75}}, \quad \ell = \sqrt{\frac{13}{25}}.$$

Its eigenvalue on $(1, 1, -1, -1)$ is

$$1 + g - h - \ell \simeq -0.05428 < 0.$$

The sign is exact: $h^2 + \ell^2 = 1 + g^2$ and $(h\ell)^2 - g^2 = 64/625 > 0$ imply $h + \ell > 1 + g$. This contradicts positivity of a Gram matrix.

The four-ray test uses the specific nonzero overlaps of the displayed realization; the full hypergraph has the different role of forcing incompatible real geometry from context incidences alone. The finite Gram-matrix gap provides a starting point for a quantitative noise analysis, but a laboratory test must justify pure preparations, rank-one tests, and the identification of equally labelled preparations and tests. Allowing unrelated preparation and measurement vectors, or higher-rank effects, changes the model being excluded. Neither this test nor the pseudocontext equality is, by itself, a device-independent or Kochen–Specker contextuality test.

More broadly, the constructions suggest investigating the operational cost of a real description: which distinctions between elementary alternatives, which projector ranks, and which completeness requirements must change when complex phases are removed? The ensemble and overlap consequences above identify concrete quantities for this investigation. Applications to steering, communication, or a quantitative resource theory remain prospects rather than consequences of the representation theorems.

VIII. CONCLUSION

The hypergraphs H_{20} and H_{21} give two exact finite separations between complex and real three-dimensional orthogonality representations. The first is governed by the field dependence of a two-projector decomposition; the second survives at the orthogonal limit and is governed by cyclic closure. Explicit algebraic vectors, complete nonexistence proofs, and exact pairwise-overlap certificates establish both results without numerical or coloring assumptions. Explicit phase-adjusted coordinates in $\mathbb{R}^6$ show at the same time that the obstruction disappears after dimension doubling, once the original three-ray contexts are no longer required to be maximal.

Physically, the examples show how a finite arrangement of complete sharp measurements can require complex amplitudes in a common three-dimensional description. In H_{20} , the forced pseudocontext permits different equal-weight pure-state resolutions of the same mixed state; in H_{21} , the obstruction survives when that local ensemble restriction disappears and resides in compatibility with the perimeter. Real simulation remains possible in a larger space, with the retained probabilities and identities determining whether projector ranks must also change. These distinctions specify which operational assumptions carry the separation and provide a finite setting in which to investigate the physical role of relative phase through preparations, measurements, and their shared experimental meaning.

Appendix A: Exact faithfulness certificates

For two nonzero representatives x, y , define their squared normalized overlap by

$$Q(x, y) = \frac{|\langle x, y \rangle|^2}{\langle x, x \rangle \langle y, y \rangle}. \quad (A1)$$

Thus $Q = 0$ exactly for orthogonal rays, while $Q = 1$ exactly for coincident rays. All calculations below are exact reductions in the algebraic fields generated by the displayed radicals and phases. The reductions and pair counts were independently cross-checked by exact symbolic arithmetic; no floating-point tolerance is used.

For H_{20} , the 12 contexts prescribe 36 distinct unordered orthogonal pairs. Substitution of Eqs. (10)–(17), followed by reduction with Eq. (11), gives $Q = 0$ for precisely those 36 pairs. For the other 154 pairs, the complete list of values and multiplicities is:

Q	mult.	Q	mult.	Q	mult.
$\frac{1}{25}$	2	$\frac{3}{10}$	8	$\frac{8}{15}$	8
$\frac{3}{50}$	4	$\frac{23}{75}$	2	$\frac{14}{25}$	8
$\frac{7}{75}$	8	$\frac{49}{150}$	8	$\frac{16}{25}$	2
$\frac{8}{75}$	8	$\frac{26}{75}$	8	$\frac{7}{10}$	8
$\frac{7}{50}$	8	$\frac{28}{75}$	8	$\frac{59}{75}$	2
$\frac{4}{25}$	8	$\frac{32}{75}$	4	$\frac{4}{5}$	16
$\frac{1}{5}$	16	$\frac{7}{15}$	8		
$\frac{6}{25}$	8	$\frac{13}{25}$	2		

(A2)

The multiplicities sum to 154. In particular,

$$\frac{1}{25} \leq Q(x, y) \leq \frac{4}{5} \quad (A3)$$

for every unprescribed pair. This proves simultaneously that no such pair is orthogonal and that no two vertices label the same ray.

For H_{21} , the 14 contexts prescribe 42 distinct unordered orthogonal pairs. Exact substitution of Eqs. (26)–(29) gives zero for precisely those pairs. The other 168 pairs split as

Q	$\frac{1}{4}$	$\frac{1}{2}$
multiplicity	84	84

(A4)

Again every unprescribed overlap lies strictly between zero and one. This completes the exact faithfulness verification used in Theorems 4 and 6.

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